

Activation functions

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Apply a declared class rule for Activation functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Activation functions. Proposed explanation: "ReLU returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace a new chain independently

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace three activations

For scores -2, 0, 2, list identity and ReLU outputs. Explain why $\text{ReLU}(2)$ is not 1.

My reasoning and evidence

6. W2 Apply a declared class rule

Use the lesson sigmoid values for -2, 0, 2. Predict classes under a ≥ 0.5 , then explain the zero-score case.

My reasoning and evidence

7. W3 Check the midpoint

Compare ReLU at inputs -2, 0, 2. Give the average of the two endpoint outputs and the output at the

midpoint; explain what this shows.

My reasoning and evidence

8. W4 Trace a new chain independently

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My reasoning and evidence

9. W5 Limit a probability claim

A model returns a sigmoid value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled activation functions evidence sheet to Image-pattern Network Investigation. Show how you can trace identity relu and sigmoid on a numerical score. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Activation functions

See what ReLU removes before the next weighted step.

Toy network $z=x-3$; $h=\max(0, z)$; $y=3h+2$. Inputs $x=1, 3, 4$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/4

ReLU hidden computation at $x=1$

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use $\max(0, \text{total}) \rightarrow \text{output } 0$

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by 0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Which original inputs cannot be distinguished by this network's final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See what ReLU removes before the next weighted step.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? y values are 2, 2, 5; $x = 1$ and $x = 3$ collapse to the same output.

1. Apply ReLU
2. Compute output
3. Compute preactivation

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compute output y values are 2, 2, 5; $x = 1$ and $x = 3$ collapse to the same output. ReLU means keep a positive value and replace a negative one with zero. The output layer receives that transformed value, not the original input. Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing ReLU with identity changes the procedure: negative intermediate values now remain available to the next layer.

Practice 2: explain the deciding evidence

Changed-case practice: Replace ReLU with identity. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. h becomes -2, 0, 1 and outputs become -4, 2, 5.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: h becomes -2, 0, 1 and outputs become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Repair a missing building block

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Representing inputs as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical inputs? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
